

## Empowering Children with Special Needs (CwSN) in Computer Science: An Interactive Hand Gestures Approach to Number Conversion Teaching

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**Abstract:** *It is essential for teachers to effectively address the diverse needs of students within the classroom environment, encompassing variations in IQ, communication skills, problem-solving abilities, and mathematical proficiency. Ensuring the inclusion of CwSN (Children with Special Needs) is imperative to prevent social division in the classroom. While the department offers accommodations such as scribes, interpreters, or extra time for examinations, these measures may not fully address the specific needs of CwSN. This study explores the efficacy of employing hand gestures as a teaching tool for CwSN in computer science, particularly in the realm of number conversions. A machine learning-based tool has been developed and implemented to provide an inclusive and accessible approach, addressing the specific needs of students with learning and intellectual disabilities. Through a mixed-methods approach, the research aims to evaluate the impact of integrating hand gestures into the instruction of number conversions. This methodology offers a simplified and interactive approach, potentially enhancing comprehension, retention, and engagement among CwSN. By focusing on the mathematical aspects of the computer science syllabus, specifically number conversion, this method aims to streamline the learning process, requiring minimal arithmetic to perform conversions without the use of pen and paper. The hand gestures method significantly improved the effectiveness of teaching number conversions to Children with Special Needs (CwSN) in comparison to traditional approaches. This approach not only boosted their participation but also facilitated easier retention of the concepts.*

**Keywords:** inclusive learning, hand gesture, number conversions, cognitive psychology, CwSN

### 1. Introduction

Children with Special Needs (CwSN) computer science learners encounter numerous challenges that can significantly impact their academic journey. These include limited accessibility of learning materials, physical barriers in learning spaces, and the availability and affordability of assistive technologies. Communication barriers and the underrepresentation of CwSN in learning materials further exacerbate these challenges. Additionally, traditional

assessment methods may not accommodate diverse learning styles, while social stigma and stereotypes can affect learners' confidence and sense of belonging. Instructors like special educators may also lack training in inclusive pedagogical approaches and resource constraints further compound these issues. Addressing these challenges is essential to creating an inclusive learning environment that caters to the diverse needs of all students, regardless of their abilities.

Fostering a positive and equitable educational experience for all students, irrespective of their backgrounds, abilities, or characteristics is important. Inclusive environments promote equal opportunities for all learners, ensuring access to the same educational resources and support (Daniel & King, 1997). By bringing together students from diverse backgrounds, cultures, and abilities, inclusivity enriches the learning experience with varied perspectives and life experiences. Engaging with diverse viewpoints enhances critical thinking, creativity, and problem-solving skills, contributing to improved learning outcomes. Additionally, inclusive learning prepares students for the globalized world by cultivating cultural competence and the ability to work effectively in diverse environments. Moreover, it promotes positive social interactions, empathy, and respect, essential for success in academic and professional settings. Recognizing and accommodating diverse learning styles ensures personalized instruction, enhancing comprehension and retention. Inclusive environments also challenge stereotypes, reduce stigma, and foster a sense of belonging for all students. Meeting legal obligations and ethical principles, inclusive practices contribute to higher levels of academic success and engagement, positively impacting students' mental health and well-being. Ultimately, inclusive learning prepares students for diverse workplaces and is a strategic investment in their educational and personal development, creating an environment where every individual can thrive and reach their full potential.

Number conversions are pivotal in computer science education, serving as a foundational concept crucial for various aspects of programming, algorithm design, and data manipulation. They are integral in several key areas like representation of information, computer architecture and assembly language, data representation and storage, networking and communication protocols, cryptography and security, digital signal processing, embedded systems programming, algorithm design and optimization, digital logic and circuit design, educational foundation etc.

The imperative for effective and inclusive teaching methods for CwSN in computer science is multifaceted, revolving around the core principles of equal opportunity, diversity, and active participation. Inclusive methods ensure equal access to education for all students, accommodating diverse learning styles and needs, thereby enhancing learning outcomes for everyone. They foster a culture of inclusion within the computer science community, addressing barriers to participation and preparing students for collaboration in a diverse workforce. Moreover, these methods align with legal and ethical standards, promote innovation and creativity, build confidence and self-efficacy, and create a positive learning environment where every student feels valued and encouraged to excel.

## **2. Literature Review**

### ***2.1. Effectiveness of Hand Gestures in Facilitating the Learning of Number Conversions***

The effectiveness of incorporating hand gestures in facilitating the learning of number conversions has garnered significant attention and interest due to its potential to enhance comprehension and engagement among learners. Research indicates that integrating hand gestures into instructional practices can serve as a powerful tool for reinforcing mathematical concepts, particularly in complex topics like number conversions. Lee (2012) examined the impact of gesture-based learning with virtual reality (GBLVR) on counting ability in three children with Cerebral Palsy (CP), revealing significant improvements in counting skills from 1 to 10 through intuitive body movement interaction with specially designed GBLVR math courses. Moreover, the kinesthetic nature of hand gestures stimulates multiple sensory modalities, catering to diverse learning styles and enhancing overall learning outcomes. Furthermore, the interactive and experiential nature of using hand gestures fosters active participation and encourages students to actively engage with the material, promoting a deeper level of understanding. Overall, the incorporation of hand gestures offers a promising approach to facilitate the learning of number conversions, providing educators with a valuable tool to support student learning and academic success in mathematics.

Learning is a multifaceted process encompassing both mental engagement and physical activity (Hannaford, 1995; Jensen, 2000). Physical movements complement cognitive functions, providing additional avenues for processing information and facilitating learning. The working memory plays a pivotal role in retaining and connecting transient information with real-world experiences and mental perceptions (Atkinson & Shiffrin, 1968; Baddeley & Hitch, 1974; Driscoll, 2005; Richey, 1986). As learners encounter real-world stimuli, they encode them, fostering effective connections between working and long-term memory. Gestures can impact how information is stored and recalled, offering alternative pathways for input and output during information processing (Cook et al., 2010). Furthermore, early research suggests that bodily engagement and environmental interactions may streamline cognitive processing by reducing the burden on memory and simplifying cognitive tasks (Donald, 1991).

An extensive examination was conducted on the role of gestures in foreign language acquisition, affirming their significant contribution to learning. Macedonia and von Kriegstein (2012) noted that the movements of the learner could enhance memory encoding, fostering deeper and more robust retention of information. This encoded information, involving motor imagery, not only penetrated deeply into memory but also improved its accessibility and durability over time. Similarly, an embodied cognition framework was adopted to explore the use of physical interactions in vocational learning. Their study made use of a Kinect technology-enabled gesture recognition system to help cognitive impaired individuals with in completing vocational tasks such as pizza making. By comparing baseline conditions without Kinect prompts to intervention conditions with Kinect-enabled prompts, Chang et al. (2013) found that the intervention significantly enhanced vocational performance. The impact of embodied interactions was studied on mnemonic performance, employing Kinect technology to provide feedback during memorization tasks. Participants who received feedback through Kinect demonstrated superior immediate and delayed recall compared to those using traditional mouse input (Chao et al., 2013). Furthermore, the effects of various embodied interactions were examined on language learning, affirming the positive influence of embodiment on the learning process. A recurring theme across these studies is the meaningful nature of body gestures and movements conveying information. Such gestures construct motor imagery that represents and metaphorically encapsulates concepts,

enabling individuals to articulate and internalize ideas through nonverbal means (Lan et al., 2018).

Body movements play a crucial role in conveying and interpreting information within its specific modality, aiding in the comprehension and representation of messages during instructional communication. Gestures are viewed as a secondary mode of communication, providing students with an additional opportunity to grasp the conveyed messages (Valenzeno et al., 2003). In their observational study on algebraic instruction, they found that gestures accompanying mathematical expressions facilitated student comprehension while also serving as indicators of the teacher's cognitive processes. These gestures not only enhanced communication during instruction but also contributed to the cognitive functions of the individuals producing them. As individuals engage in the learning process, they engage in internal communication, which is integral to their own thinking and cognitive activities. Body movements can aid in this "inner" communication, enriching and supporting cognitive processes. (Alibali & Nathan, 2007).

After analyzing empirical studies in neuroscience and education, Kiefer and Trumpp (2012) observed that actions associated with words or concepts could trigger memory recall, potentially aiding in retrieval. Additionally, they noted that sensory-motor experiences contributed to the development of more robust conceptual frameworks compared to experiences lacking such involvement, due to the meaningful interactions with the subject matter. Moreover, they emphasized that embodied experiences, particularly when closely related to learning objectives, have the potential to strengthen learning outcomes. The research underscores the significance of sensory and motor interactions in the learning process, suggesting that appropriate engagement with these modalities is needed for the human cognition optimal development (Kiefer & Trumpp, 2012).

In an early study, Cook and Goldin-Meadow (2006) investigated how young learners utilized gestures to tackle mathematical problems involving the addition of three integral numbers with both equivalent and nonequivalent addends on two sides of the equation. Their findings proved the efficiency of gestures in facilitating learning such mathematical content. Similarly, studies were conducted examining how gestures aid in mathematical learning (Broaders et al., 2007) echoing the results of Cook and Goldin-Meadow. They observed that gestures were utilized by learners to solve and process mathematical ideas and concepts. These studies identified several reasons for how effective gesture in learning reduces cognitive load. Gestures directly influenced the storage of new representations in memory, prompted the construction of imaginative representations for later recall, and established connections between real-world problems and learners' mental models. It was found that the embodiment of physical conceptions and actions in school mathematics emphasized activities such as playing with shapes, collecting them, pointing and counting, sharing, and measuring (Tall, 2008). Furthermore, a positive impact of embodied interactions was demonstrated on the learning of angular momentum, noting the activation of students' sensorimotor brain regions while reasoning about the topic (Kontra et al., 2015).

While perspectives on embodiment may vary, the majority trace their roots back to psychological processes influenced by the body (Glenberg, 2010). During the teaching and learning of mathematics, particularly regarding equations and their graphical representations, highlighting how embodied interactions play a role in the multimodal transfer of knowledge.

Arzarello et al. (2009) found that body movements offered alternative avenues for organizing and embodying information, especially complex concepts that might not be entirely expressible through speech alone. For instance, using gestures to illustrate tangent lines on a graph facilitated learners' comprehension of related equations and calculations. This study illustrated how gestures served as a modality for both expressing and absorbing intricate concepts within the thinking process. Similarly, Yoon et al. (2011) conducted a case study on math learning by observing participants using a combination of gestures and speech. They noted how participants used hand movements to metaphorically represent abstract mathematical concepts such as downhill slopes symbolizing negative gradients. Nathan and Walkington (2017), as well as Walkington et al. (2014), argued that dynamic gestures explained various states of mathematical ideas and integrated motor experiences into the mental encoding process. Ozcelik and Sengul (2012) developed a virtual 3D environment utilizing Kinect technology to teach vector concepts, allowing participants to control avatars and draw 3D vectors using arm movements in a virtual classroom. While this was an exploratory endeavor, it demonstrated the potential of embodying complex STEM concepts through computer-mediated 3D simulations. Furthermore, Parmar et al. (2016) revealed a positive attitude among middle school students in computational thinking while exploring embodied interactions within an immersive virtual learning environment.

Johnson-Glenberg et al. (2014) investigated that learners within a computer-mediated immersive multimedia environment engaged with learning interventions using body movements and gestures. These movements and gestures were intricately designed to symbolize and represent the learning content. For example, circling a designated object in the multimedia scenario was accomplished by making circles with the arm, reflecting specific physics concepts. The researchers conducted empirical studies focusing on learning in two STEM subjects: titration in chemistry and disease transmission. They argued that their learning module stimulated various neuronal pathways in the learner's motor system, with movements tailored to match the knowledge being taught. Results from their research favored learning with embodied interactions over traditional instruction methods. In a follow-up investigation into the immersive learning of concepts concerning electric fields, including principles like Coulomb's law, Johnson-Glenberg and Megowan-Romanowicz (2017) extended their exploration of embodiment levels. They pinpointed three crucial elements—sensorimotor engagement, congruency between gestures and content, and the level of immersion—as critical factors shaping the efficacy of embodiment in the learning process. Different combinations of these facets were found to produce varied learning effects.

Lindgren et al. (2016) guided 7th-grade students to physically enact the trajectory of an asteroid, utilizing their comprehension of gravity as a foundation. The cohort employing embodied learning techniques showcased notable advancements in learning and displayed heightened levels of engagement and positive attitudes compared to the control group, which utilized a desktop simulation. Similarly, Hung et al. (2014) employed Kinect technology to track hand gestures of 5th-grade students, allowing them to control the positions of light beams and candles to grasp fundamental optics concepts. The embodied learning group showed superior learning performance compared to the control group, which interacted with the simulation using a mouse and keyboard.

Skulmowski et al. (2016) investigated the impact of embodied learning by providing

university students with a 3D plastic model of a human heart equipped with motion tracking sensors. Manipulating the model with their hands and arms, students engaged with corresponding learning materials displayed on a screen in real-time. The embodied learning group exhibited significantly better learning outcomes compared to their counterparts using a mouse. In Kang et al.'s study (2018), undergraduate students utilized gestures symbolizing linear and exponential operations to engage with an earthquake simulation system, resulting in improved comprehension of numerical concepts. Li et al. (2019) focused on pointing gestures in multimedia instructional materials designed to teach undergraduates about the nervous system. Pointing gestures directed at specific parts of the material significantly improved learning outcomes compared to general pointing or non-pointing gestures. Additionally, Price (2020) explored how bodily movements can aid in understanding geometry concepts such as space and coordinates.

Amelia et al. (2016) studied middle-school students learning about the connections between linear equations and corresponding graphs during mathematics instruction, and found that while teachers often used gestures to refer to representations, students learned less when the teacher referred to equations in gesture, compared to when she did not. This result suggested a potential trade-off in attention to visual representations and highlighting the need for a nuanced understanding of the role of teachers' gestures in student comprehension and learning. Donadello et al. (2023) provided an overview of the role of gestures in education. They emphasized their effectiveness in aiding second language learning, providing inclusive education for students with disabilities, and teaching complex concepts in science and math, thus highlighting gestures as a potent yet often overlooked means of communication and learning.

Roth (2001) aimed to fill the gap in educational research by reviewing existing literature across anthropology, linguistics, psychology, and education to highlight the significance of gestures in knowing, learning, and teaching, and proposes pertinent focus questions for further educational research. Chang et al. (2023) developed a hand gesture recognition system to enhance human-computer interactions for the speech-impaired community, whose communication relied solely on gestures, using a two-phase algorithm involving Region of Interest Segmentation and a Convolutional Neural Network mode. They demonstrated a 10 percent increase in accuracy compared to systems without image segmentation, underlining the necessity for improved representation of the speech-impaired community in human-computer interaction research. Solomon et al. (2018) completed a qualitative study observing a high school computer science class and found that students' gestures during coding activities served as external representations of their understanding. Proficient students exhibited more general gestures indicating comprehension, while struggling students relied on specific pointing gestures for each code line, suggesting that teachers could use students' gestures as a formative assessment tool to gauge learning.

Previous research indicates that embodied interactions can yield positive outcomes in learning activities. Body movements play a pivotal role in augmenting cognitive processes, reinforcing content within working memory, and facilitating the encoding of perception and observation. Early inquiries into incorporating body movements for learning STEM-related subjects primarily centered on traditional face-to-face classroom environments and were largely tailored towards younger learners (Alibali & Nathan, 2007, 2012; Arzarello et al., 2009; Edwards, 2009; Goldin-Meadow et al., 2009; Reynolds & Reeve, 2001). Recent research has shown a

growing interest in utilizing computer-mediated environments and emerging body tracking technology to explore how body movements relate to learning across diverse disciplines. (Hung & Chen, 2018; Johnson-Glenberg et al., 2014; Johnson-Glenberg & Megowan- Romanowicz, 2017; Lan et al., 2019). These investigations have tackled STEM-related topics and principles such as friction, gravity, electromagnetism, human physiology, and geometric concepts (Kang et al., 2018; Lindgren et al., 2016; Magana & Balachandran, 2017a, 2017b; Merkouris et al., 2019; Price et al., 2020; Skulmowski et al., 2016). However, a shared limitation across these studies is their primary focus on conceptual comprehension and retention. For instance, Lindgren et al. (2016) explored gravitational concepts through an embodied analogy, devoid of numerical computations, where bodily movements symbolized the trajectory of an asteroid. Likewise, Hung et al. (2014) associated arm gestures with fundamental principles of optics, evaluating elementary students' grasp of these ideas.

In this research, our goal is to delve deeper into the efficacy of embodied interactions, particularly focusing on gestures, within the context of informal computer science education at the higher secondary level. Our aim is to explore how CwSN learners not only grasp concepts and principles but also apply them to perform calculations, specifically in the realm of number conversions involving decimal and binary systems.

## ***2.2 Existing Method of Number Conversions***

Traditional teaching methods for number conversions often involve step-by-step procedures and examples to help students understand the process. Here are common approaches used in traditional teaching for number conversions, particularly focusing on decimal to binary and vice versa.

### ***A. Decimal to binary conversion***

#### ***1). Division remainder method***

- a. Divide the decimal number by 2.
- b. Record the remainder (it will be a binary digit).
- c. Repeat the process with the quotient until the quotient is 0.
- c. Read the binary digits in reverse order to get the binary equivalent.

#### ***2). Repeated division***

- a. Divide the decimal number by 2 and record the remainder.
- b. Divide the quotient by 2 and record the remainder.
- c. Continue this process until quotient is 0.
- d. Read the remainders in reverse order to get the binary equivalent.

#### ***3). Doubling method***

- a. Start with a binary place value of 1.
- b. Double the place value successively while checking if it is less than or equal to the decimal number.
- c. If the doubled value is less than or equal to the decimal number, write down a “1” and subtract the doubled value.
- d. Repeat the process until the decimal number becomes 0.

## ***B. Binary to decimal conversion***

### *1). Place value method*

- a. Write down the binary number.
- b. Assign place values to each binary digit from right to left (1, 2, 4, 8, 16 etc.)
- c. Multiply each binary digit by its corresponding place value.
- d. Sum up the results to get the decimal equivalent.

### *2). Doubling and adding method*

- a. Start with a decimal value of 0.
- b. Double the current decimal value and add the next binary digit.
- c. Repeat this process for each binary digit.

### *3). Powers of 2*

- a. Write down the binary number.
- b. Assign powers of 2 to each binary digit, starting from  $2^0$ .
- c. Multiply each binary digit by its corresponding power of 2.
- d. Sum up the results to get the decimal equivalent.

## ***2.3. Studies on the Use of Gestures in Education***

An experimental study was conducted to explore the efficacy of using Unity3D and Kinect V2 for teaching binary and decimal number conversions to novice adult learners through embodied interactions. The research by Xu & Fe (2021) compared an experimental group, which utilized body movements for learning, with a control group employing conventional mouse and keyboard interactions. The findings revealed that while embodied interactions enhanced conceptual learning, they did not significantly improve knowledge retention and application when compared to traditional methods. This highlighted the importance of considering implementation and intensity when designing effective learning systems with novel technologies.

### **3. Methods**

We focused on examining whether hand gestures method could lead to greater knowledge acquisition and application in math calculations about numeric systems, when compared to traditional methods, in a virtual learning environment for higher secondary CwSN.

This research aims to assess the effectiveness of using hand gestures in teaching number conversions to CwSN computer science learners by reviewing existing literature, identifying gaps, and defining research questions. As a result, research questions were developed to address the impact of hand gestures on teaching number conversions, their effects on CwSN learners, and potential challenges and benefits.

#### ***3.1. Procedures and Participants***

We have selected a diverse group of CwSN computer science learners from Malappuram district of Kerala, India. These constitute 60 per cent of the total number of CwSN in computer science (first year and second year) in Malappuram district of Department of Higher Secondary Education of Kerala, India.

We ensured the participants represented intellectual disability (ID) and specific learning disability (SLD) as represented in Table 1. Participants are selected based on characteristics such as age, computer science knowledge, and disabilities, with informed consent obtained. An experimental intervention was designed, incorporating hand gestures into a learning program tailored to the needs of CwSN, while a control group received standard instruction. Pre-intervention assessments measured baseline understanding of number conversions, and post-intervention assessments gauged the impact of the hand gestures intervention.

We divided the participants into two groups: one exposed to the hand gestures method (intervention group) and another using traditional number conversions using teaching methods (control group). We randomized participants to control for potential bias. The learning materials encompassed fundamental concepts including numeric weight, number conversions involving binary and decimal. Within the intervention group, learners engaged with the content interactively by physically interacting with a widescreen display. Through gestures, they dynamically navigated through the learning materials and exercises, visually embodying the concepts being taught. For instance, to grasp the concept of “weight” in relation to digit position, learners raised or lowered specific fingers to represent the weight associated with each digit’s position. This interactive approach aimed to deepen understanding by integrating physical movement with abstract concepts, fostering a multisensory learning experience. These gesture interactions, involving movements and positioning, served to establish cognitive connections between digit positions and their respective weights. Further insights and discussions regarding the effectiveness and implications of these interactive methods were elaborated upon in the Discussion section, offering a deeper understanding of the pedagogical approach employed.

In the control group, participants followed conventional number conversion methods akin to those typically employed in a classroom setting, utilizing pen and paper. Conversely, in the intervention group, participants were randomly assigned to utilize hand gestures as a method to

learn identical concepts.

**Table 1**

*Information About the Participants*

| Parameters                      | Intervention group<br>(Hand gesture method) | Control group<br>(Traditional method) |
|---------------------------------|---------------------------------------------|---------------------------------------|
| Gender                          |                                             |                                       |
| Male                            | 487                                         | 389                                   |
| Female                          | 355                                         | 305                                   |
| Average age                     | 16.3                                        | 16.5                                  |
| Educational background          | First year                                  | First year                            |
| Stream/Combination              |                                             |                                       |
| Computer Science                | 502                                         | 438                                   |
| Computer Application (Commerce) | 340                                         | 256                                   |
| Disability type                 |                                             |                                       |
| Intellectual Disability         | 498                                         | 391                                   |
| Specific Learning Disability    | 356                                         | 303                                   |

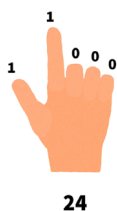
Each participant underwent a pretest on the learning content before commencing the intervention under either condition. Subsequently, an immediate posttest was administered. Three days later, participants were requested to complete a delayed posttest to assess retention and comprehension over time. We selected ten percent (1536) of the total students with ID or SLD (2561) who either enrolled in plus one computer science or plus one computer application courses. Out of a total of 1536 adult participants, 1536 completed the study intervention and both tests. Among the 1536 participants, 100% underwent the study, with half initially assigned to the intervention group and the others to the control group, after which their roles were interchanged. Pre and post tests were conducted for both groups. Notably, 90 percent of the intervention group successfully retained the method taught during the intervention and were able to achieve the correct final answer. Conversely, less than 20 percent of the control group managed to achieve the correct answer using the traditional method.

### **3.2. Hand Gesture Method**

This method uses hand gestures to perform calculations for converting decimal to binary and vice-versa. The method explained below uses left hand (facing towards you), each finger from little finger to thumb represents a bit as shown in Figure 1 for binary to decimal conversion. Fingers represent weight in ascending powers of 2 as shown in Figure 2 in decimal to binary conversions. In all scenarios provided, the conversion assumes a 5-bit binary number for binary to decimal conversion, and for binary to decimal conversion, the decimal numbers involved are restricted to values below 32.

**Figure 1**

*Bit Values Associated with Fingers*



**Figure 2**

*Weights Associated with Fingers*



### **3.2.1. Binary to decimal conversion**

Step 1: Identify the number of bits to determine the number of fingers needed.

Step 2: Represent the binary number as a hand gesture. (Closed finger = 0, raised finger = 1).

Step 3: Add the numbers represented by the raised fingers only.

For example, let's convert the binary number 10001 to decimal.

Number of bits: 5.

Add the numbers in the raised fingers as shown in Figure 3 (i.e.,  $16 + 1 = 17$ )

Therefore, the decimal equivalent of 10001 is 17.

**Figure 3**

*Binary to Decimal Conversion for 10001*



### **3.2.2. Decimal to binary conversion**

Step 1: Identify the number of bits to determine the number of fingers needed.

Step 2: Represent the decimal number as a sum of values in fingers.

Step 3: Raise the needed fingers and close the unwanted ones as shown in Figure 4.

Step 4: Raised finger = 1, closed finger = 0 as seen in Figure 5.

**Figure 4**

*Hand Gesture Representation of Decimal 24*



For example, let us convert the decimal number 24 to binary (see Figure 5). The decimal number 24 is below 32, so we need 5 bits (i.e., 5 fingers)

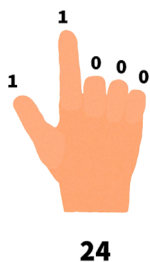
Number of bits: 5.

Decimal number: 24

Therefore, the binary equivalent of 24 is 11000.

**Figure 5**

*Hand Gesture Representation of Decimal 24 in Binary*



#### **4. Implementation and Tools Used**

Hand gestures method is done using Python language along with Mediapipe, Opencv, and ScikitLearn using the concept of Machine Learning. We have done this in 4 phases.

##### ***Phase 1 - Collecting the Image Data***

In Phase 1, we utilized the webcam of a laptop to capture 100 images for each gesture by moving hands in front of the camera. This process aimed to enhance the accuracy of results by ensuring variability in hand gestures. The captured images were organized and stored in respective folders, with each folder representing a specific gesture. Considering the scenario of 5-bit binary numbers, we created 32 folders, each labeled with the corresponding binary number (from 0 to 31), and stored 100 images for each binary number. This systematic approach facilitated the creation of a diverse and comprehensive dataset for subsequent phases of the project.

##### ***Phase 2 - Creating the Data Set***

In Phase 2, we focused on extracting the spatial patterns and geometry of the hand and fingers to create a dataset for gesture recognition. We specifically considered the left hand and its four fingers, aligning with the scenario of a 4-bit binary number representation. By analyzing the spatial arrangement and geometrical features of the hand and fingers, we generated a dataset that captures the essential characteristics necessary for accurate gesture recognition. This dataset was then serialized and stored as a pickle file, ensuring easy accessibility and efficient utilization in subsequent phases of the project.

##### ***Phase 3 - Training the Classifier***

In Phase 3, we utilized the dataset generated in Phase 2 to train a random forest classifier for gesture recognition. By feeding the spatial patterns and geometrical features of hand gestures into the classifier, we trained it to accurately classify different gestures. After training, the classifier successfully created a model capable of recognizing hand gestures with high accuracy. Evaluation of the model revealed that it achieved a remarkable accuracy score of 100%, correctly classifying all samples in the dataset. This exceptional performance demonstrates the effectiveness of the random forest classifier in accurately identifying hand gestures based on the extracted features. Additionally, the classifier provides the capability to generate classification results for new input data, thereby facilitating real-time gesture recognition applications.

##### ***Phase 4 - Testing the Model***

In Phase 4, the trained model underwent testing by providing sample hand gesture inputs to obtain the actual output. During testing, the model successfully recognized and classified the hand gestures, yielding the correct binary value for the presented gestures. However, important to note that the accuracy of the classification results was contingent upon the quality and correctness of the images provided during Phase 1. High-quality and accurately labeled images in the dataset

contribute to better training and subsequently improved performance during testing. Therefore, to ensure reliable and accurate results, maintaining consistency and precision in image data collection and labeling processes from the initial phases of the project was important.

Firstly, the use of hand gestures can offer a tangible and interactive way for learners to conceptualize abstract mathematical concepts, such as number conversions. By associating specific hand movements with numerical operations, learners can visualize and internalize these concepts more effectively, potentially leading to improved comprehension and retention. Additionally, the kinesthetic nature of hand gestures engages multiple sensory modalities, catering to diverse learning styles and enhancing overall learning outcomes. This approach may be particularly beneficial for learners who struggle with traditional teaching methods or who have special educational needs. Moreover, incorporating hand gestures into the learning process can promote active participation and engagement among students, fostering a dynamic and interactive classroom environment. This active involvement can lead to increased motivation and interest in the subject matter, ultimately enhancing the overall learning experience. However, it is important to acknowledge that the effectiveness of the hand gestures method may vary depending on factors such as the complexity of the mathematical concepts being taught, the proficiency level of the learners, and the quality of instruction. Additionally, the implementation of this method may require adequate training and support for educators to ensure its successful integration into the curriculum.

## **8. Discussions**

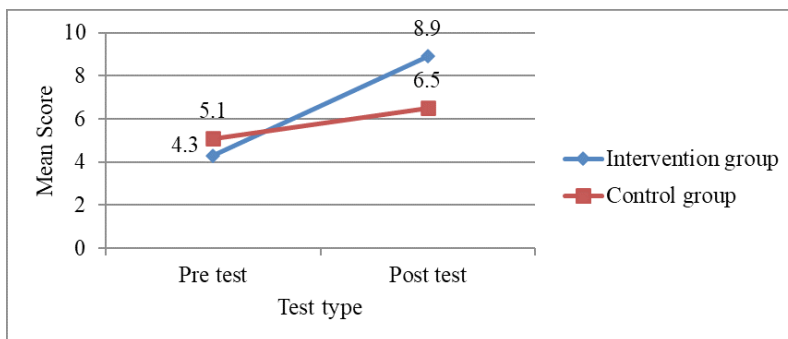
In discussing the research, exploring strengths and limitations of the research study, along with its potential implications for teaching and learning are important.

We have developed a series of 10 standardized number conversion tasks with varying levels of complexity, encompassing questions that require conceptual clarity and problem-solving skills. These tasks cover different number systems, including binary and decimal representations. Basics of number systems and problems involving number conversions were taught to both groups before they underwent the pretest. Participants were given instructions about pretest (like the total number questions, type of questions, marks and time) and given opportunity to take a pretest.

Scores were recorded based on their conceptual understanding and problem-solving abilities. In the next week, after instruction of hand gesture method, a posttest was administered for intervention group and the control group, after a revision of traditional method, underwent the posttest. The results (see Figures 6, 7 and 8) showed that the intervention group had more effective learning and memory retention skills to solve the problems and for better memory retention. In this case, the hand gesture method proved more effective. Surprisingly, the learners with difficulty in learning and intellectual disability were able to solve the problems and can understand the concepts in a better way.

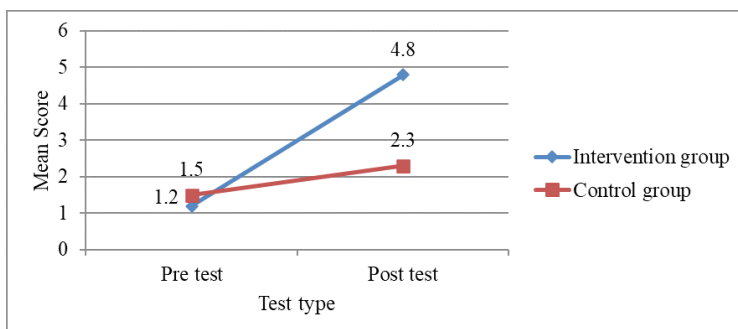
**Figure 6**

*Mean Scores of Both Groups (10 questions, 10 marks)*



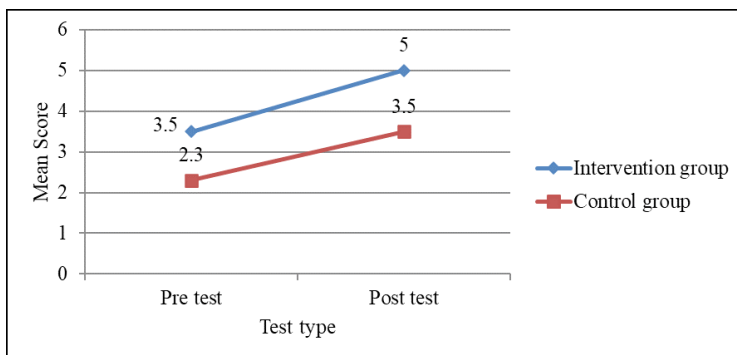
**Figure 7**

*Mean Scores of Both Groups (5 Concept Questions, 5 Marks)*



**Figure 8**

*Mean Scores of Both Groups (5 Problem Questions, 5 Marks)*



#### **4.1. Feedback**

The positive feedback from students regarding the hand gestures method has significant implications for inclusive teaching practices in computer science as shown in Table 2.

**Table 2**

*Student Feedback on Hand Gesture Method*

| <b>Feedback parameter</b> | <b>Number of responses</b> | <b>Positive response</b> |
|---------------------------|----------------------------|--------------------------|
| 1.Engagement              | 1516                       | 80%                      |
| 2.Understanding           | 1516                       | 76%                      |
| 3.Enjoyment               | 1438                       | 75%                      |
| 4.Effectiveness           | 1516                       | 89%                      |
| 5.Clarity                 | 1516                       | 93%                      |
| 6.Comfort                 | 1438                       | 81%                      |
| 7.Confidence              | 1438                       | 84%                      |
| 8.Interest                | 1516                       | 83%                      |
| 9.Ease of Use             | 1516                       | 96%                      |
| 10.Overall                | 1516                       | 91%                      |

By incorporating interactive and multisensory learning approaches such as hand gestures, educators can create a more inclusive learning environment that caters to diverse learning styles and abilities. This approach not only enhances engagement and understanding among students with intellectual or learning disabilities but also benefits all learners by promoting active participation and deeper comprehension of complex concepts.

Moreover, the success of the hand gestures method highlights the importance of leveraging innovative instructional strategies and technologies to create accessible and inclusive learning experiences in computer science education. By embracing inclusive teaching practices, educators can foster a supportive and empowering learning environment that enables all students to thrive and succeed in their academic pursuits.

#### **4.2. Limitations**

Several constraints and challenges were encountered during the research. These are explained further in accordance with their categories.

**Accessibility issues:** Ensuring accessibility for CwSN learners posed a significant challenge, particularly in designing and implementing hand gesture-based learning materials that catered to various disabilities.

**Technical complexity:** Developing the hand gesture recognition system and integrating it with the learning platform required advanced technical expertise and resources, adding complexity to the project.

**Participant recruitment:** Identifying and recruiting participants with intellectual or learning disabilities, who met the study criteria proved challenging, as it required collaboration with specialized educational institutions and coordination with caregivers or guardians.

**Ethical considerations:** Adhering to ethical guidelines and obtaining informed consent from participants, especially those with disabilities, necessitated careful planning and sensitivity to their rights and privacy.

**Training and support:** Providing adequate training and support for both participants and special educators in using the hand gesture-based learning system posed logistical challenges, particularly in ensuring consistent implementation across different settings.

**Data collection and analysis:** Collecting and analyzing data from CwSN learners presented unique challenges, including variability in response times, communication styles, and cognitive abilities, which required careful consideration in data interpretation. The support of the special educators in this aspect is worth mentioning.

**Time and resource constraints:** Limited time and resources posed constraints on the scope and scale of the research, impacting factors such as sample size, intervention duration, and the breadth of data collection.

Despite these challenges, the research project ultimately provided valuable insights into the effectiveness and feasibility of using hand gestures as a method of learning number conversions for CwSN (ID and SLD) computer science learners. Addressing these constraints and challenges required interdisciplinary collaboration, innovative problem-solving, and a commitment to inclusive research practices.

### ***4.3. Future Research Recommendations***

Further exploration and refinement of the hand gestures method for learning number conversions among CwSN computer science learners offer promising avenues for future research and development:

**Customization for specific disabilities:** Tailoring hand gesture-based learning materials to address the unique needs and abilities of different types of disabilities, such as visual impairments, motor disabilities, or cognitive impairments, could enhance accessibility and effectiveness.

**Adaptive learning technologies:** Developing adaptive learning technologies that dynamically adjust hand gesture-based instructions based on individual learner profiles and progress could optimize engagement and learning outcomes for CwSN.

**Longitudinal studies:** Conducting longitudinal studies to assess the long-term effectiveness and retention of knowledge gained through hand gesture-based learning methods could provide valuable insights into the sustainability of learning outcomes over time.

**Multimodal approaches:** Exploring the integration of hand gestures with other sensory modalities, such as auditory cues or tactile feedback, could enhance the multisensory learning

experience and accommodate a broader range of learning preferences and abilities.

**Teacher training and support:** Providing comprehensive training and support for special educators in implementing hand gesture-based teaching methods effectively, including strategies for inclusive instruction and classroom management, is essential for scaling up the approach in diverse educational settings.

**Cross-disciplinary collaboration:** Collaborating across disciplines, including computer science, education, psychology, and assistive technology, could facilitate the development of innovative solutions and best practices for integrating hand gestures into inclusive teaching practices.

**Real-world application:** Exploring real-world applications of hand gesture-based learning methods beyond number conversions, such as programming concepts or problem-solving skills, could extend the utility and relevance of the approach in computer science education.

## **5. Conclusion**

Hand gestures solving number conversions proved very effective among the CwSN learners as compared to the traditional methods. This method guaranteed more effective participation from them and helped them to retain the concepts much easily.

The study found that employing hand gestures as a teaching method for number conversions among CwSN computer science learners effectively enhanced conceptual learning. Learners exhibited heightened engagement and enthusiasm while utilizing hand gestures to interact with the learning materials, reflecting a positive response to the instructional approach. This multisensory learning experience allowed participants to integrate physical movement with abstract concepts, thereby deepening their understanding of the material. Additionally, the intervention group demonstrated significantly higher retention rates of the learning method taught during the intervention compared to the control group, indicating the efficacy of hand gestures in fostering knowledge retention and application. However, challenges such as implementation and intensity were identified, underscoring the need for careful design and planning when employing this instructional approach. Overall, the findings suggested that the hand gestures method of learning number conversions held promise as an effective and engaging instructional strategy for CwSN computer science learners, potentially leading to improved learning outcomes and retention.

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### **Declaration Statement**

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### **Ethics Approval: Yes**